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[22CSBS107]

FOURTH SEMESTER B.Tech EXAMINATION, MAY-2024

DATABASE MANAGEMENT SYSTEMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Mention different categories of SQL Languages	L1	CO1	**02**
	b.	List down the integrity Constraints in Relational Database.	L1	CO2	**02**
	c.	List out the desirable properties of decomposition	L1	CO3	**02**
	d.	List out the states of transaction.	L1	CO4	**02**
	e.	Define SQL Injection.	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Explain roles and responsibilities of a DBA.	L2	CO1	**05**
		OR			
	b.	Briefly explain about the following Data Models. **i.** Hierarchical Data model **ii.** Network Data model	L2	CO1	**05**
3.	**a.**	Draw the ER Diagram to a model Online Book store.	L3	CO2	**05**
		OR			
	b.	Explain the Aggregate functions in MySQL with examples.	L2	CO2	**05**
4.	**a.**	Let's suppose we have a set of attributes as S: {A, B, C, D, E, F, G, H} and functional dependencies are A -> BC E -> A B -> CFH CH -> G F -> EG. Identify, how many candidate keys present in the above table?	L3	CO3	**05**
		OR			
	b.	Briefly explain about different types of Functional Dependency with an example.	L2	CO3	**05**
5.	**a.**	Explain in detail about B Tree index files.	L2	CO4	**05**
		OR			
	b.	State query optimization and explain its two forms.	L2	CO4	**05**

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6.	**a.**	Identify, how do intrusion detection systems work? Discuss the different types of Intrusion Detection.	L2	CO5	**05**
		OR			
	b.	Describe about Data warehousing.	L2	CO5	**05**
7.	**a.**	Consider a relation schema R (A , B , C , D) with the functional dependencies A → B and C → D. Identify whether the decomposition of R into R1 (A , B) and R2 (C , D) is lossless or lossy?	L2	CO3	**05**
		OR			
	b.	Explain briefly about Evaluation of relational algebra expression.	L2	CO3	**05**
		PART-C			**6 X 10=60**
8.	**a.**	Explain briefly about Architecture of the Database management systems with neat diagrams.	L2	CO1	**10**
		OR			
	b.	Consider the following table and write the query for the following questions.	L3	CO1	**10**

Table: Employee

Id	last_name	first_name	city	reg_year	Age
125	Anderson	Mary	Seattle	2021	25
101	Jones	Chris	Bellevue	2020	32
104	Thompson	Lora	Seattle	2020	43
113	Farrell	Brian	Bellevue	2021	48
108	Smith	Jeff	Bellevue	2020	60
115	Peters	Anna	Bellevue	2020	72
127	Patterson	John	Bellevue	2020	31

i. Write SQL query to Create and insert the values.

ii. *Write SQL query to* display the persons whose age between 35 and 70.

iii. Write SQL query to Select the persons who are residing at Bellevue city

iv. Write SQL query to Select All those Last names ends with "on"

v. Write SQL query to Set unique and not null constraints for id column

9.	**a.**	Explain the basic Relational algebra operations with the symbols used and an example for each.	L2	CO2	**10**
		OR			
	b.	List out the different types of joins in SQL? Explain with an example.	L2	CO2	**10**
10.	**a.**	Explain the 2NF, 3NF and BCNF normal forms with examples.	L2	CO3	**10**
		OR			
	b.	Discuss the steps involved in query processing.	L2	CO3	**10**
11.	**a.**	Define Serializability. Explain the types of serializability with examples	L2	CO4	**10**
		OR			
	b.	Explain how Two-phase Locking Protocol used in transactions.	L1	CO4	**10**
12.	**a.**	Distinguish between DAC and MAC of Database Access Control.	L4	CO5	**10**
		OR			
	b.	Discuss in detail about the architecture of the distributed databases	L2	CO5	**10**
13.	**a.**	Explain briefly about basic and additional rules of Armstrong Axioms of Functional Dependency.	L2	CO3	**10**
		OR			
	b.	Explain in detail about the different types of Indices in Database management systems.	L2	CO4	**10**